Numerical evaluation of Tweedie probability functions

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12 November 2026

Introduction

Data types: Continuous data

- **Continuous on real line** (e.g., altitude): $Y \in \mathbb{R}$.
  - Model using a normal distribution (standard regression).
- **Positive continuous** (e.g., crop yields): $Y \in \mathbb{R}_+$.
  - Model using gamma distribution (a generalized linear model; GLM).
- **Proportions** (e.g., cloud cover): $Y \in [0, 1]$.
  - Model using beta distribution.
- **Directions** (e.g., movements of animals): $Y \in [0, 2\pi)$.
  - Model using von Mises distribution.



Data types: Discrete data

- **Discrete** (e.g., counts of insects): $Y \in \mathbb{N}$
  - Model using Poisson distribution (a GLM).
- **Binomial counts** (e.g., seed germination): $Y \in \{0, 1, 2, \ldots, m\}$.
  - Model using binomial distribution (a GLM).



Data types: Mixed data

- **Continuous data with exact zeros** (e.g., rainfall): $Y \in \{0\} \cup \mathbb{R}_+$.
  - What probability model...?



- Rainfall (e.g., Yunus et al. (2017), Hasan and Dunn (2010); Dunn (2004)).
- Fisheries: Catch per unit effort (CPUE) (e.g., Candy (2004); Dunn (2009)).
- Root length density of trees (e.g., Dunn and Smyth (2005); Silva et al. (1999)).
- Insurance payouts (e.g., Haberman and Renshaw (1996)).
- Alcohol consumption in teenagers (e.g., Gilchrist and Drinkwater (1999)).
- Survival analysis (e.g., Aalen (1992)).
- Transcriptomics (e.g., Mallick et al. (2022); Wu et al. (2023)).

Mixed data: models

Continuous data with exact zeros can appear for different reasons:



Censoring

- A detection or measurement limitation.
- *Example*: Pollutant or chemical concentrations below the instrument's detection limit.



Structure

- The structure of the data naturally leads to mixed data.
- *Example*: Daily rainfall: day with no rain, amount exactly 0; otherwise a positive continuous amount.

PG models: model development

- **Number of precipitation events in any month** N.
  - N has a *Poisson* distribution: $N \sim \text{Pois}(\lambda)$ (note: N could be zero).
- **Precipitation amounts per event** when rain falls R.
  - Rainfall event i receives rainfall R_i; *gamma* distribution: $\text{Gam}(-\alpha, \gamma)$.
- ***Total* monthly precipitation**, Y.
  - Poisson sum of gamma random variables:

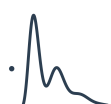
$$Y = \begin{cases} 0 & \text{if } N = 0; \\ R_1 + R_2 + \cdots + R_N & \text{if } N > 0. \end{cases}$$



PG models: development

The probability distribution of a **Poisson sum of gamma variables**:

- can be deduced by working with cumulant generating functions.
- is complicated: generally **no closed form** apart from well-known special cases:
  - Normal distributions.
  - Gamma distributions.
  - Poisson distributions.
- **Poisson–gamma models** belong to the **exponential dispersion model** (EDM) family, so fit the GLM framework (Dunn and Smyth (2018)).



PG models

- *All* EDMs have **mean** $\mathrm{E}[Y] = \mu$.
- *All* EDMs have **variance** $\mathrm{var}[Y] = \phi V(\mu)$, where:
  - $\phi > 0$ is some constant (the **dispersion parameter**);
  - $V(\mu)$ is the **variance function**, uniquely defining the EDM distribution.

Tweedie (power-variance) models

Distribution	**Var fn** $V(\mu)$
Normal distribution	$V(\mu) = 1$
Poisson distribution, if $\phi = 1$	$V(\mu) = \mu$
Gamma distribution	$V(\mu) = \mu^2$
Inverse Gaussian	$V(\mu) = \mu^3$
'Equivalent' to $y^* = 1/y$ transform in regression	$V(\mu) = \mu^4$

Tweedie (power-variance) models

Distribution	**Var fn** $V(\mu) = \mu^\xi$	ξ
Normal distribution	$V(\mu) = 1$	0
Poisson distribution, if $\phi = 1$	$V(\mu) = \mu$	1
Gamma distribution	$V(\mu) = \mu^2$	2
Inverse Gaussian	$V(\mu) = \mu^3$	3
'Equivalent' to $y^* = 1/y$ transform in regression	$V(\mu) = \mu^4$	4
Tweedie (power-variance)	$V(\mu) = \mu^\xi$	$\notin (0, 1)$
Poisson–gamma	$V(\mu) = \mu^\xi$	$\in (1, 2)$

Tweedie (power-variance) models

Range of ξ	**Distribution Type**
$\xi \leq 0$	Continuous on $(-\infty, \infty)$; e.g., Normal ($\xi = 0$)
$0 < \xi < 1$	**No distributions exist**
$\xi = 1$	Non-negative discrete; e.g., Poisson ($\xi = 1$ and $\phi = 1$)
$1 < \xi < 2$	Mixed on $y \geq 0$ (mass at $y = 0$); Poisson–gamma distributions
$\xi \geq 2$	Continuous on $(0, \infty)$ e.g.: Gamma ($\xi = 2$); inv. Gauss ($\xi = 3$)

Power-variance densities: examples

Tweedie PDF examples

Tweedie distributions

The Tweedie family is a three-parameter family of distributions:

- $\mu > 0$ (the **mean**);
- $\phi > 0$ (the **dispersion parameter**);
- ξ (the **index parameter**: *which* Tweedie distribution, sometimes denoted as p).

- The estimation of μ follows standard GLM algorithms... if ξ is known.
- Tweedie GLMs can be fitted in Genstat, R, SPSS, etc.



Tweedie distributions: evaluation

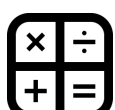
- Two important functions for describing a distribution:
  - Probability density function (PDF): how probability is **distributed**.
  - Distribution function (DF): how probability **accumulates**.

Use **PDF** $f_\xi(y)$ to:

- **estimate** a value for ξ (and hence fitting a model...).
- compute maximum likelihood estimates.
- compute the AIC.

Use **CDF** $F_\xi(y)$ to:

- **evaluate models**: computing quantiles and p-values.
- **compute quantile residuals**.
- generate **random numbers**.



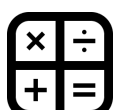
Tweedie distributions: evaluation

Tweedie PDFs must be, in general, evaluated numerically:

- Saddlepoint *approximation*: accurate for small dispersion ϕ.
- Infinite series (Dunn and Smyth 2005).
- Infinite oscillating integrals (Dunn and Smyth 2008).

Tweedie DFs must be, in general, evaluated numerically (Dunn and Smyth 2027) :

- Saddlepoint *approximation*: accurate for small dispersion ϕ.
- Direct integration of the PDF.
- Infinite series for $1 < \xi < 2$.
- Infinite oscillating integrals.



PDF evaluation: series

PDF evaluation: series

- For $1 < \xi < 2$: PDF is written in terms of Wright's generalized Bessel function $W(y, \phi, \xi)$:

$$\sum_{j=1}^{\infty} \frac{y^{-j(2-\xi)/(1-\xi)}(\xi-1)^{(2-\xi)/(1-\xi)j}}{\phi^{j(1-(2-\xi)/(1-\xi))}(2-\xi)^j\, j!\, \Gamma\big(-j(2-\xi)/(1-\xi)\big)}$$

- A similar expression exists for $\xi > 2$.
- Messy, but evaluating in principle is easy: sum over many terms$\sim j$.
- Sometimes the terms are a *long way* from $j = 1$.
- Sometimes a very large number of terms are needed.
- So **which** terms, and **how many** terms, must be summed?

PDF evaluation: series strategy

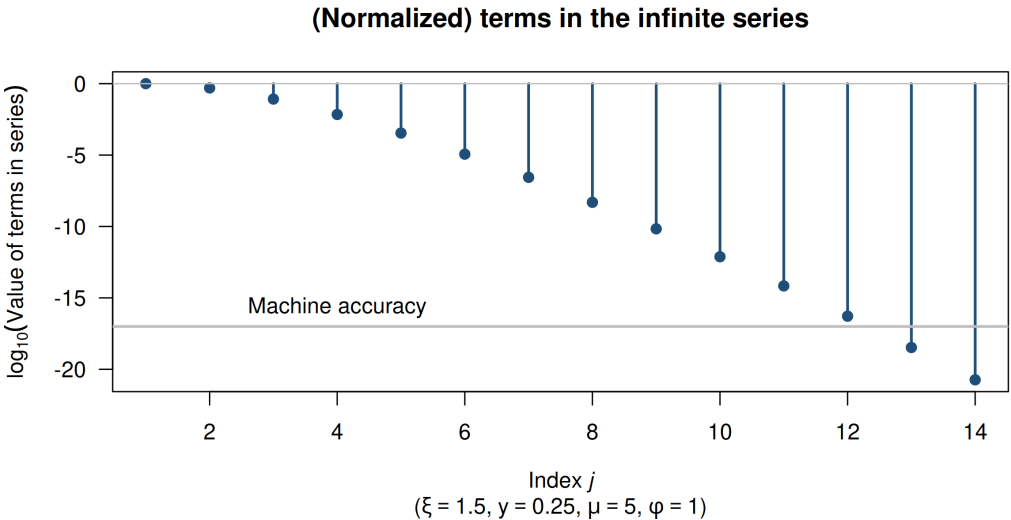
We directly sum the infinite series, only over the necessary terms:

- Using Stirling's approximation for the gamma function, plus other approximations...
- ... then treat the index of summation j as continuous...
- ... and differentiate to find where the terms reach a maximum in terms of j.

Then: sum terms either side of the maximum until the contributions are negligible.



PDF evaluation: series

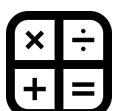


PDF evaluation: series

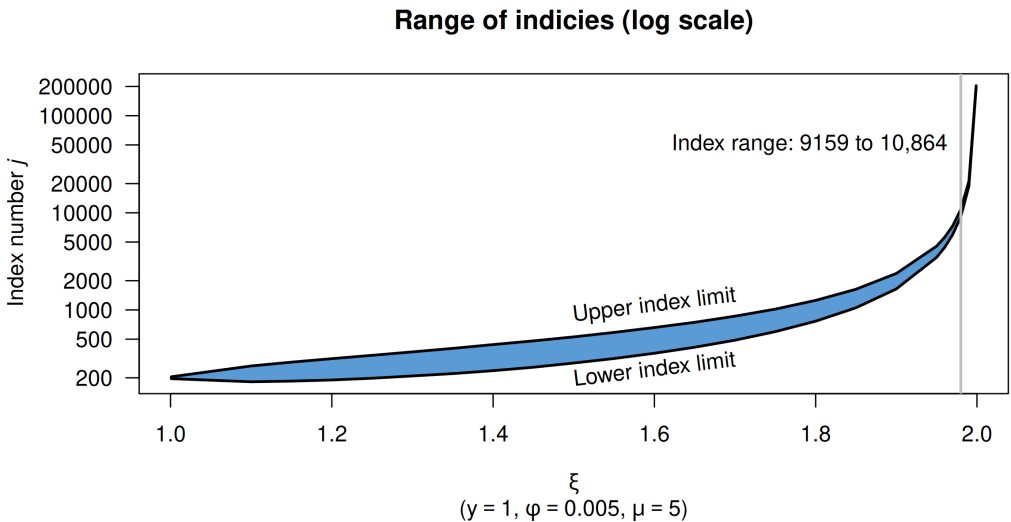
Challenges:

- Sometimes, the **number of terms** required is in the thousands.
- Sometimes, the **contributions** don't start until $j = 100\,000+$.

- Sometimes, the numerical evaluation is tricky (e.g., factorials involving index$\sim j$).



PDF evaluation: series



PDF evaluation: series performance

- The series method works well:
  - only summing positive terms, so machine accuracy is achievable.
- However, the number of necessary terms grows without bound as $\xi \to 2$ and as $\Pr(Y = 0) \to 0$.
- In these cases, the method becomes very slow.



PDF evaluation: Fourier inversion

PDF evaluation: Fourier inversion

- Tweedie distribution have no closed form density...
- ... but have relatively simple moment generating functions (MGFs):

$$M_Y(t) = \mathrm{E}\left[e^{tY}\right] = \int_{-\infty}^{\infty} e^{ty} f(y)\, dy \quad \text{(by definition)}$$
$$= \exp\left\{\frac{\kappa(\theta + t\phi) - \kappa(\theta)}{\phi}\right\},$$

where

$$\theta = \frac{\mu^{1-\xi} - 1}{1-\xi}, \qquad \kappa(\theta) = \frac{\mu^{2-\xi} - 1}{2-\xi}.$$

- $\xi = 1$ (Poisson) and $\xi = 2$ (Gamma) have a closed form: focus is $\xi \neq 1, 2$.

PDF evaluation: Fourier inversion

- Fourier inversion of the **Moment Generating Function** $M_Y(t)$:

$$\begin{aligned} f_Y(y) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp[K_Y(it) - ity]\, dt \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp[k(t)]\, dt \end{aligned}$$

where $i = \sqrt{-1}$, and $K_Y(t) = \log M_Y(t)$.

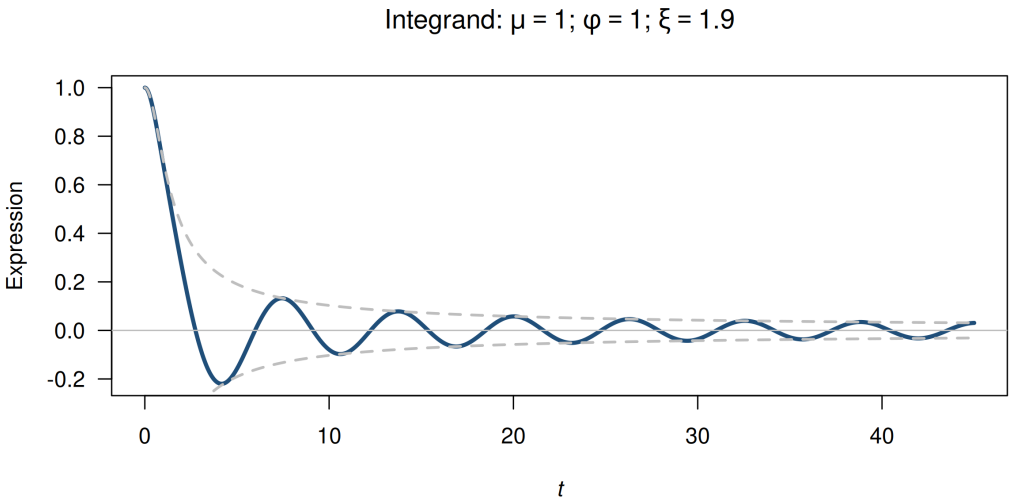
PDF evaluation: Fourier inversion

- Extracting the **real components**:

$$f_Y(y) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp[\Re k(t)] \cos \Im k(it)\, dt,$$

- This is an infinite, oscillating integral:
  - **exponentially decaying** (i.e., $\exp[\Re k(t)]$); and
  - **oscillating** (i.e., $\cos \Im k(it)$) function.

PDF evaluation: Fourier inversion



PDF evaluation: Fourier inversion

Areas in integration

PDF evaluation: Fourier inversion

$$f_Y(y) = \frac{1}{\pi} \int_0^\infty \exp[\Re k(t)] \cos \Im k(it)\, dt,$$

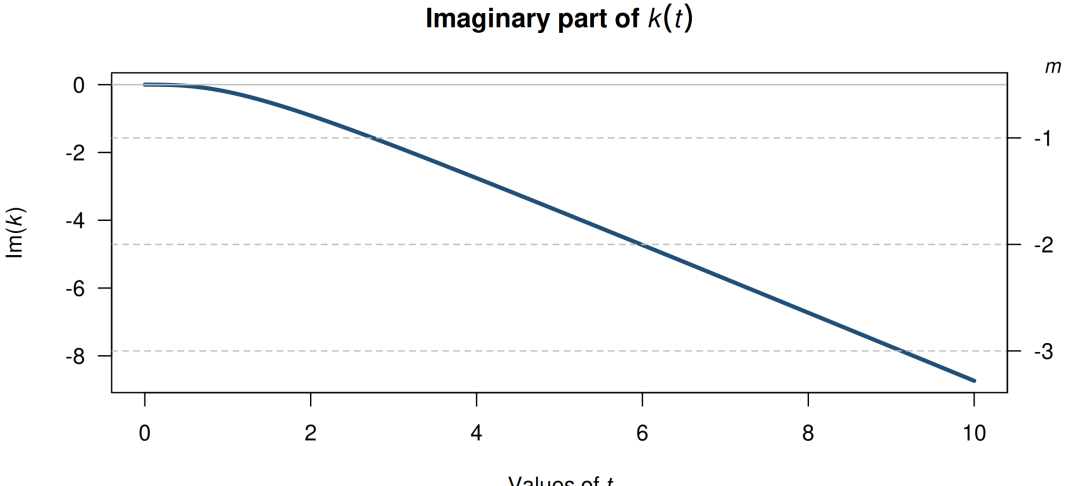
- The integrand zeros occur when

$$\cos \Im k(t) = 0 \implies k(t) = \frac{\pi}{2} + m\pi \quad \text{for integer } m.$$

- To find the zeros, the behaviour of $k(t)$ needs to be studied...

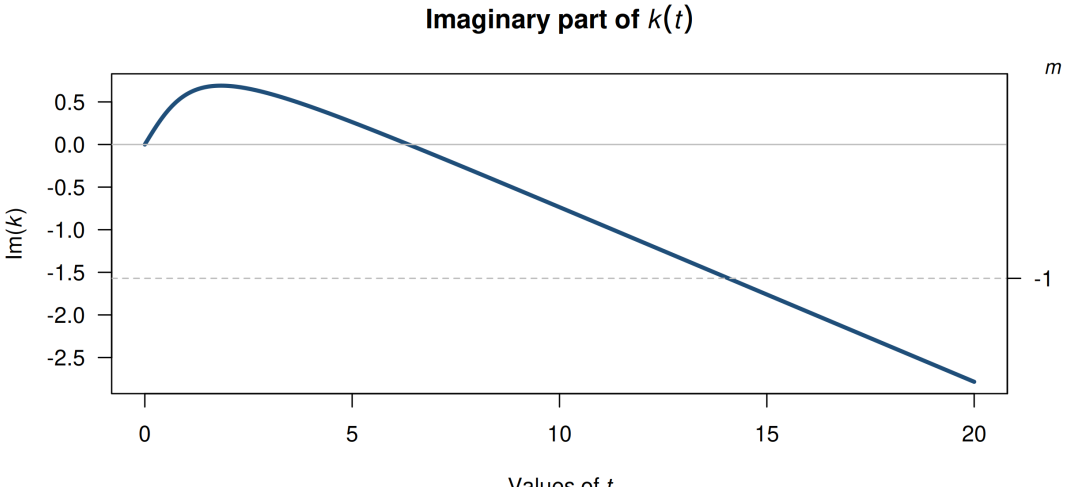
PDF evaluation: Fourier inversion

$\frac{d}{dt}\Im k(t) < 0$, so solve $k(t) = \frac{\pi}{2} + m\pi$ for $m = -1, -2, \ldots$



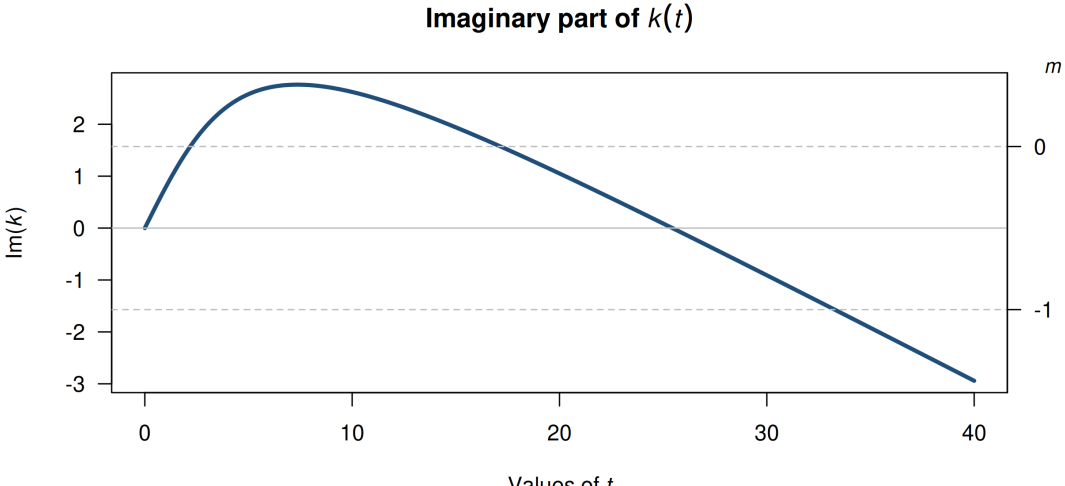
PDF evaluation: Fourier inversion

$\frac{d}{dt}\Im k(t) > 0$; max. value $< \pi/2$, so solve $k(t) = \frac{\pi}{2} + m\pi$ for $m = -1, -2, ...$

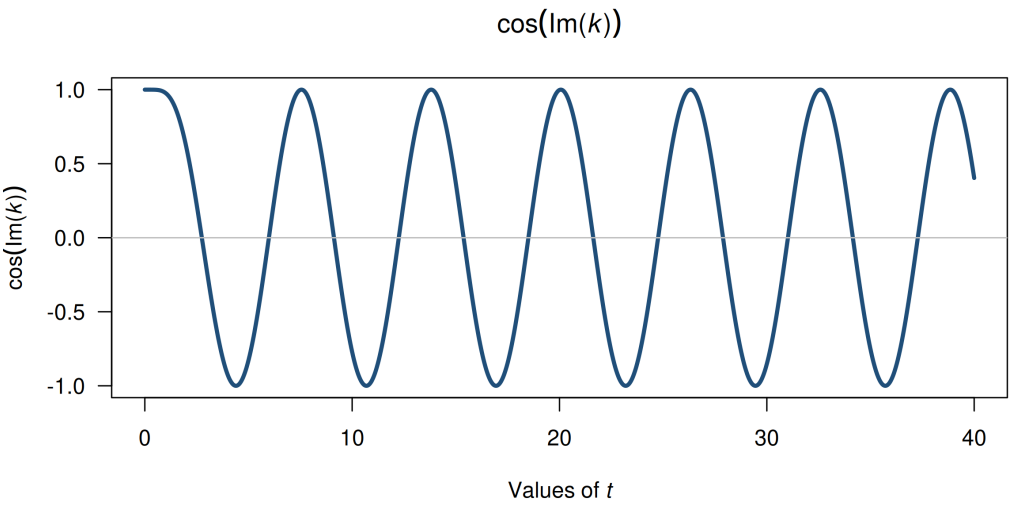


PDF evaluation: Fourier inversion

$\frac{d}{dt}\Im k(t) > 0$; max. value $\geq \pi/2$, so solve $k(t) = \frac{\pi}{2} + m\pi$ for $m = 0, 0-1, ...$



PDF evaluation: Fourier inversion



PDF evaluation: Fourier inversion



In principle, the algorithm is simple:

- Find the zeros of the integrand...
- Find the area of each integrand (numerical quadrature)...
- Sum the areas.



Challenges:

1. Accumulated rounding error summing almost-equal, positive and negative areas.
2. Integrands are not always nicely behaved...

PDF evaluation: Fourier inversion

Challenge 1: Rounding error

- Accumulated rounding error summing almost-equal, positive and negative areas.
- **Numerical acceleration** techniques speed the convergence.
- Many acceleration techniques available, depending on the type of decay behaviour.
- We use Sidi's W-transform (Sidi (1980), Sidi (1982)).



PDF evaluation: acceleration techniques

Areas as series

PDF evaluation: Fourier inversion

Challenge 2: Integrand behaviour

- Integrands are not always this nicely behaved...
- Lots of mathematical understanding of the integrand helps.
- Using an identity to simplify the integrand sometimes helps:

$$f_\xi(y;\mu,\phi) = c\, f_\xi(cy; c\mu, c^{2-\xi}\phi)$$

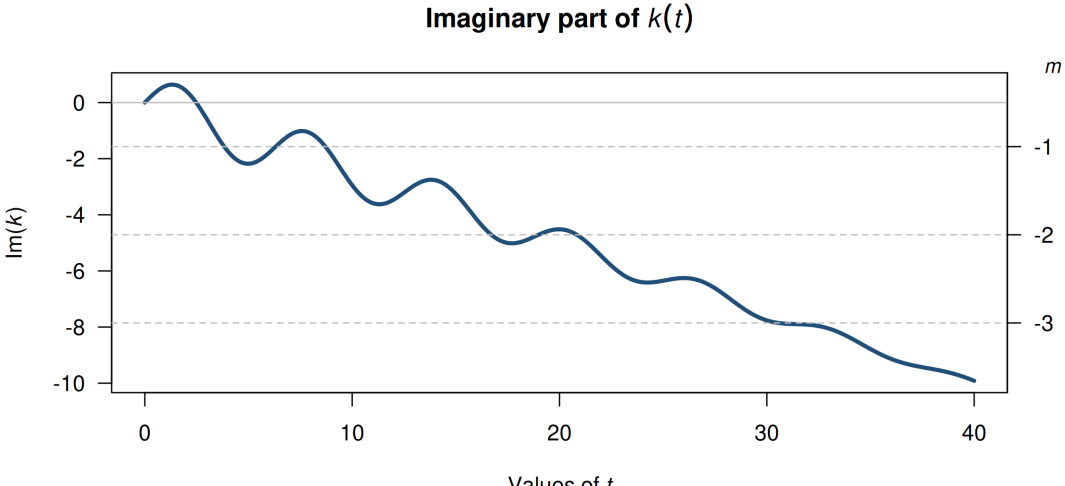
  for a constant $c > 0$.
- For example: with $c = 1/\mu$

$$f_\xi(y;\mu,\phi) = \tfrac{1}{\mu}\, f_\xi(y/\mu; 1, c^{2-\xi}\phi).$$



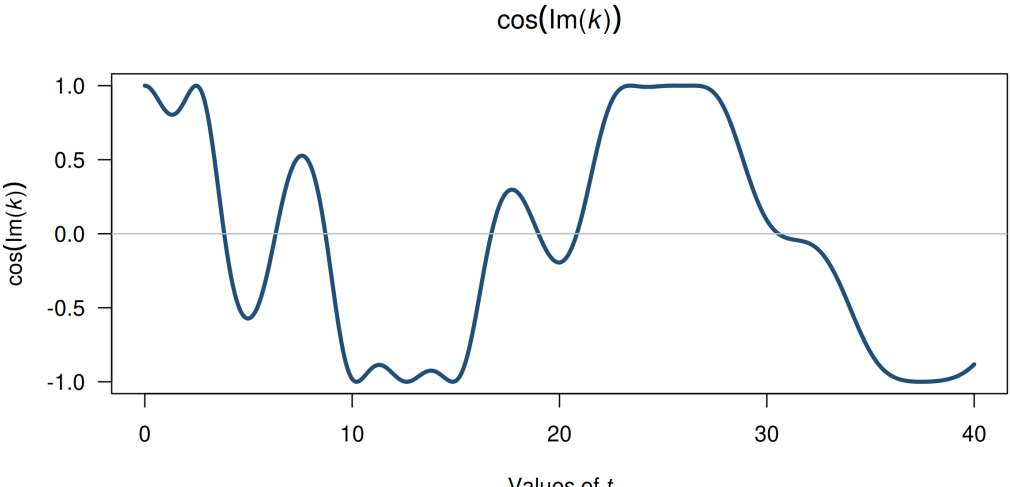
PDF evaluation: finding the zeros

Solve $k(t) = \frac{\pi}{2} + m\pi$ for $m = -1, -1, -1, -2, -2, -2, -3, ...$



PDF evaluation: finding the zeros

Oscillations tend to cosine as $t \to \infty$, but this may take a while...



PDF evaluation: Fourier performance

- The Fourier inversion method works well where the series does not:
  - as $\xi \to 2$ and as $\Pr(Y = 0) \to 0$.
- Using the series and Fourier inversion techniques together essentially cover the entire sample space for accurate evaluation.



DF evaluation

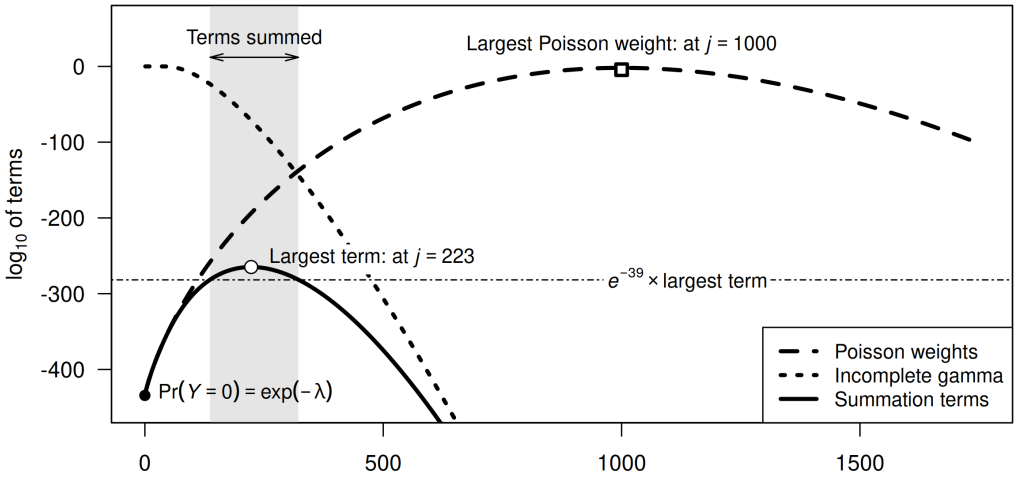
DF Evaluation strategies

- A series expression is available for $1 < \xi < 2$; none for $\xi > 2$.
- For $1 < \xi < 2$: smallest relative error with evaluating series.
- For $\xi > 2$: smallest relative error requires different strategies:
  - Body of distribution: requires Fourier inversion
  - Tails: require direct integration.
  - Small dispersion: requires saddlepoint approximation.



DF evaluation: series for $1 < \xi < 2$

- DF series evaluation has different strategy than PDF:



DF evaluation: strategies for $\xi > 2$

- Fourier inversion has similar ideas (zero finding; acceleration).
  - Integrand is a damped sin-curve rather than a damped cos-curve.
- High relative accuracy is required, and often in the extreme tails of the distribution.
  - Fourier inversion has good *absolute* accuracy, but not good *relative* accuracy in the extreme tails.
  - Direct integration is useful here.

DF evaluation: integration for $\xi > 2$

- Direct integration of the density:

$$F_p(y) = \Pr(Y = 0) + \int_0^y f_p(t)\,dt \quad \text{(by definition)}$$
$$= \Pr(Y = 0) + f_p(y) \int_0^y \frac{f_p(t)}{f_p(y)}\,dt.$$

- Dividing by $f_p(y)$ makes the integrand equal to one at $t = y$, so the integral is not extremely small even when the probability is tiny.
- Computing the result as $\log f_p(y) + \log \int(\cdot)$ prevents underflow.
- The values of $f_P(t)$ come from the numerical methods given earlier:
  - **Computationally costly**.
  - Requires very accurate density evaluations!
- Gauss–Laguerre is efficient for evaluating the integrals in the tails however.

DF evaluation: saddlepoint approx.

- A cheap computational method is the Lugannani–Rice saddlepoint approximation to the distribution function [LugannaniRice1980; Daniels1954]:
- The saddlepoint, solving $K'(\hat{t}) = y$, is $\hat{t} = (y^{1-p} - \mu^{1-p})/\{\phi(1-p)\}$, and

$$F(y) \approx \Phi(\hat{w}) + \varphi(\hat{w}) \left(\frac{1}{\hat{w}} - \frac{1}{\hat{u}} \right), \quad \hat{w} = \operatorname{sgn}(y-\mu)\sqrt{d(y,\mu)/\phi}, \quad \hat{u} = \hat{t}\sqrt{\phi V(y)},$$

  where $d(y, \mu)$ is the unit deviance, and Φ and φ are the standard normal distribution and density functions.
- At very small dispersion (i.e., very small ϕ), the approximation is very accurate, just where the inversion for $p > 2$ loses accuracy.

DF evaluation: rel. accuracy: $p > 2$

Excellent relative accuracy except for extreme cases ($\rho = \mu^{2-\xi}/(\phi(2-\xi))$)

Regime	**Method**	**Worst rel. error**
$\rho \leq 10^4$	Inversion (tail: integration when prob $< 10^{-5}$)	4×10^{-10}
$10^4 < \rho \leq 10^5$	Inversion (tail: integration when prob $< 10^{-2}$)	2×10^{-9}
$10^5 < \rho \leq 10^7$	Integration; saddlepoint in body	2×10^{-8}
$\rho > 10^7$	Saddlepoint	2×10^{-8}
$y \ll \mu$; $\rho \leq 10^4$	Inversion	5×10^{-1} with warning

Example: the data

- Total July rainfall at Quilpie:

	Year	Rain	SOI	Phase
1	1921	38.4	2.7	2
2	1922	0.0	2.0	5
3	1923	0.0	-10.7	3
4	1924	24.4	6.9	2
5	1925	0.0	-12.5	3
6	1926	9.1	-1.0	4

- Small data set: $n = 68$ July totals.
- Exactly zero rainfall recorded for 20 months (29%).



Example: Quiplie rainfall

- The model is

$$\begin{cases} Y_i \sim \text{Tweedie}(\mu_i; \xi) & \text{(random)} \\ \log \mu_i = \mathbf{X}\vec{\beta} & \text{(systematic)} \end{cases}$$

- **Randomness**: data mixed: use a Poisson–gamma probability model.
- **Structure**: log-link function.
  - Ensures predicted means always positive values.
  - Suits the data.
  - The effect of the covariates x_j on μ is *multiplicative* not *additive*.
  - As a covariate: use the SOI.

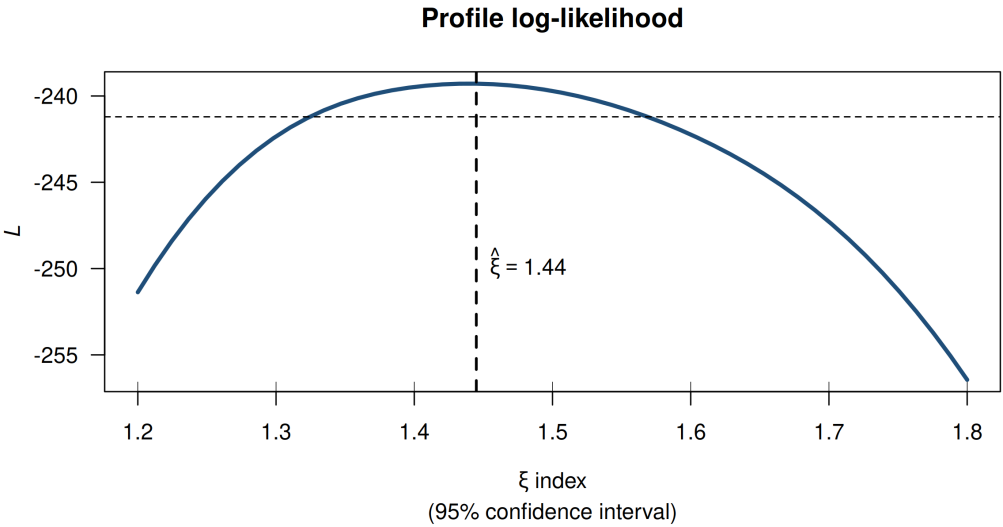
Example: process

Fitting **process** like fitting other GLMs, with some idiosyncrasies.

1. Determine *which* Tweedie distribution is appropriate (i.e., the value of ξ).
2. Fit the GLM, usually using a *log* link.
   - Usual Wald tests (t-tests) and likelihood ratio tests (F-tests) can be used.
   - AIC needs full likelihood calculations.
3. Examine diagnostics (using quantile residuals if possible) to evaluate model.

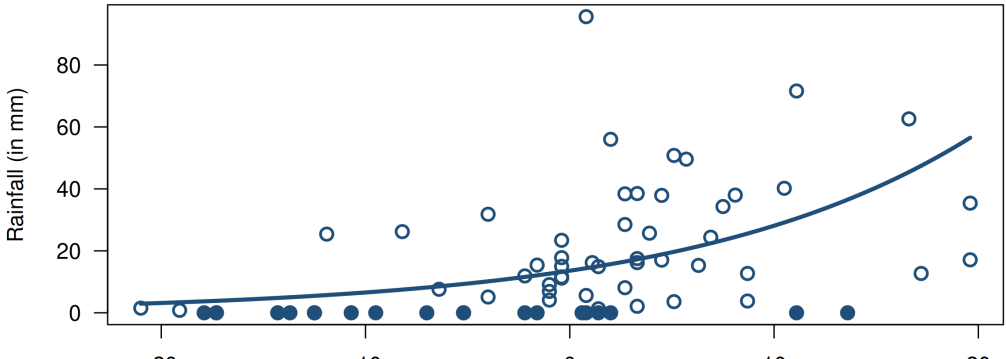


Example: find the value of ξ



Example: the model

$$\begin{cases} Y_i \sim \text{Tweedie}(\mu_i; \xi = 1.4) & \text{(random)} \\ \log \mu_i = 2.61 + (0.073 \times \text{SOI}) & \text{(systematic)} \end{cases}$$



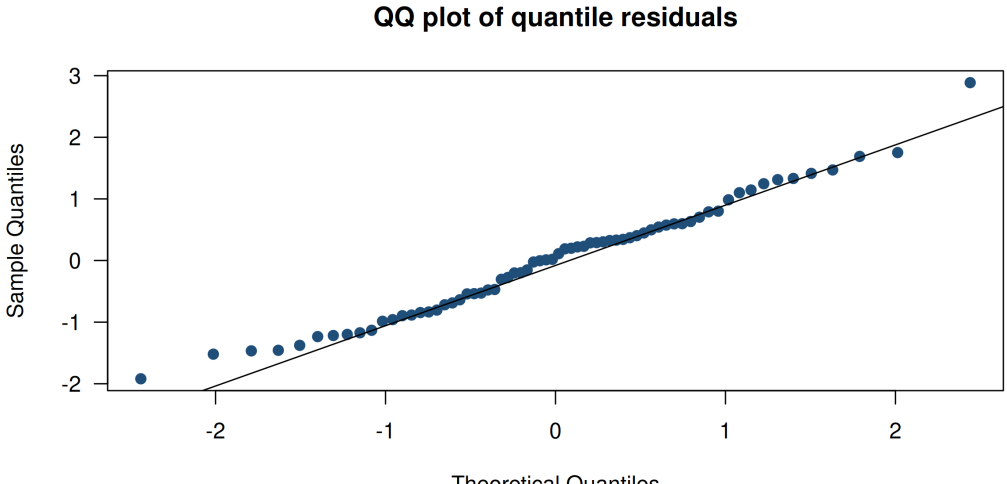
Example: model evaluation

Variable	Coefficients	Standard error	t-score	P-value
Intercept	2.61	0.152	17.1	< 0.001
SOI	0.073	0.0165	4.42	< 0.001

Model	Deviance	DF	Resid deviance	F-score	P-value
Intercept only		67	617.44		
+ SOI	120.63	66	496.81	17.09	< 0.001

Example: diagnostic analysis

Using quantile residuals (Dunn and Smyth (1996)):



Concluding

Accuracy

- No closed form exists (in general) for the PDF or DF.
- So no 'gold standard' exists for **checking the accuracy and precision** of the algorithms.
- But, the algorithms:
  - give accurate results for the closed-form cases ($\xi = 2$ and $\xi = 3$);
  - convergence occurs to machine accuracy where possible;
  - give sensible answers;
  - produce similar answers when using different methods.



Packaging

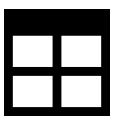
Algorithms prepared and packaged for use in the `tweedie` package (Dunn 2026).

- Freely available on CRAN.
- Coded in R and FORTRAN.
- **Over 1 million downloads**.



Combined and faster approach

- We can switch between methods, and use the most accurate method for any given (y, ξ, μ, ϕ) combinations.
- We also have a faster method, using accurate interpolation of stored precise values (to be published).
- Much faster, but loses some accuracy in some limiting cases.



The future

Parts of the parameters space (i.e., (y, ξ, μ, ϕ) combinations) are still challenging for accurate evaluation.

- Other evaluation method are possible, which may work well and/or faster in parts of parameter space.
- We have plans for exploring these...



References I

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